

# CAN-FD Powertrain Bus Timing & Protocol Test Report

Dual Bitrate Characterization, Bus-Off, and E2E Profile 01 per ISO 11898-1/2

## 1. CAN-FD Physical Layer Dual Bitrate Timing

- Test Case TC-NET-051: Oscilloscope bus timing analysis confirmed 500.0 kbps arbitration and 5.00 Mbps data phase with 75.0% sampling rate.
- Bus propagation delay measured: 142 ns over 25m topology.

## 2. CAN Bus-Off Recovery State Machine

- Test Case TC-NET-052: Fault injection bus-off test: node resumed transmission on CAN bus at 94.6 ms following error frame burst. Verification passed.

## 3. AUTOSAR E2E Profile 01 Checksum Verification

- Test Case TC-NET-053: E2E Profile 01 verified on BMS status messages (IDs 0x100 to 0x108). Inverter and DC-DC message IDs (0x200 to 0x208) also verified.